

OVULATION

Headings

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PROCESS / CASCADE OF OVULATION

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1. DETERMINATION OF BASAL BODY TEMPERATURE (BBT)
2. DETERMINATION OF HORMONAL EXCRETION IN URINE
3. DETERMINATION OF HORMONAL LEVEL IN PLASMA
4. ULTRASOUND SCANNING
5. CERVICAL MUCUS PATTERN

- **Definition** : The physiological process by which a mature Graafian follicle ruptures and the ovum is released into the abdominal cavity.
- **Timing** : Occurs characteristically on the **14th day** of a normal 28-day menstrual cycle.
- **Number of Ova** : Only **one ovum** is typically released from one of the ovaries per cycle.
- **Primary Trigger** : **Luteinizing Hormone (LH)** is primarily responsible for inducing ovulation (*LH surge*).

PROCESS / CASCADE OF OVULATION

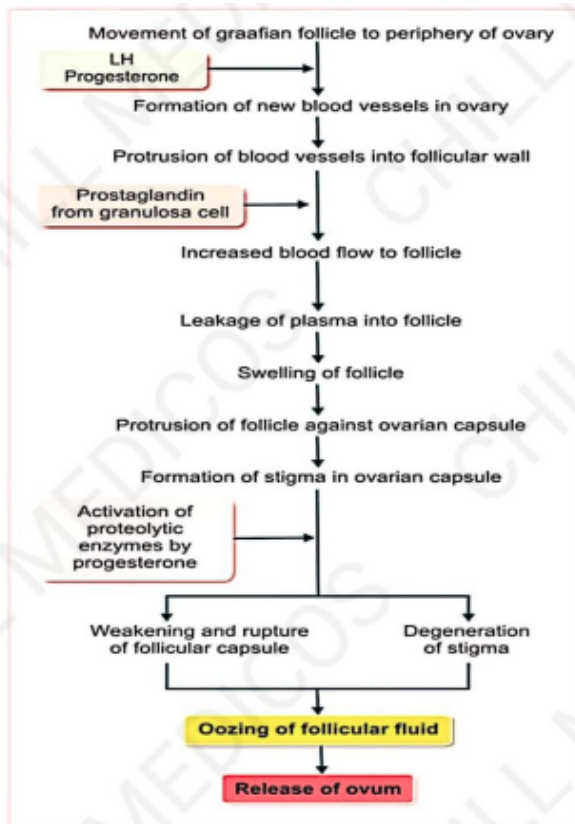
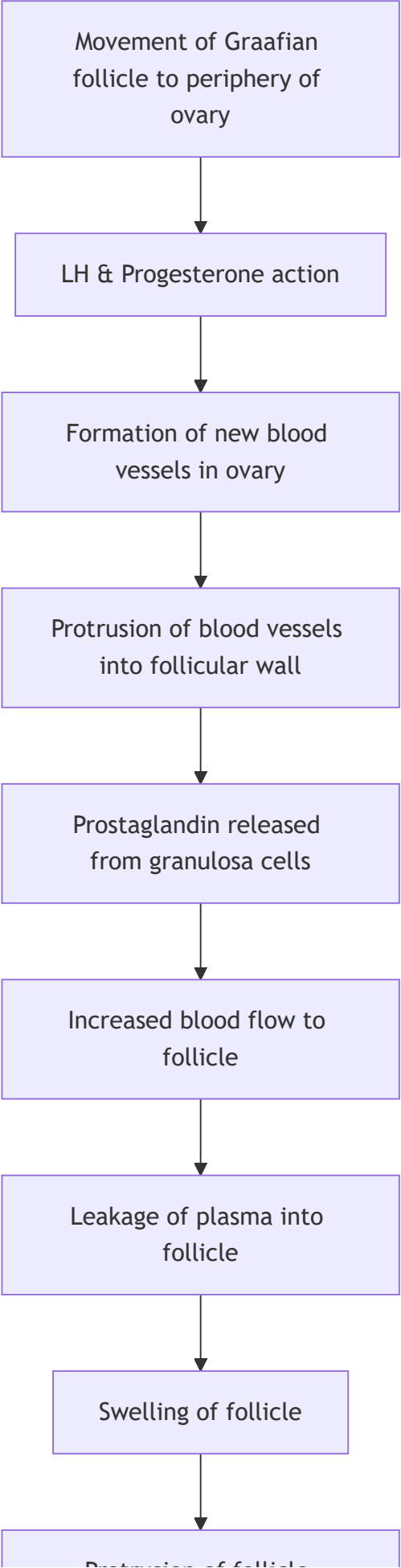
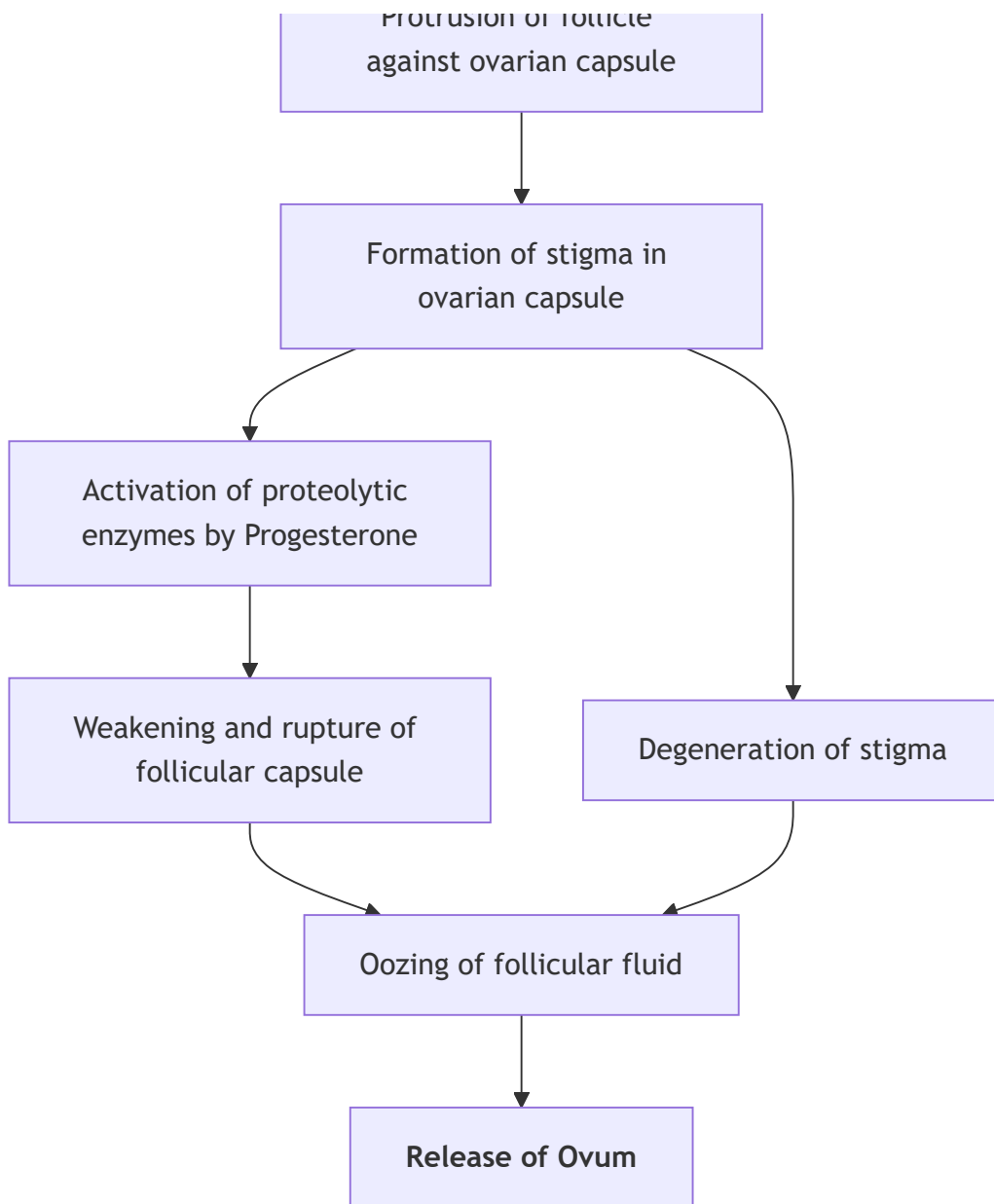


FIGURE 81.1: Process of ovulation





DETERMINATION OF OVULATION TIME

- **5 Methods to Determine Ovulation Time :—**

- a. Determination of Basal Body Temperature (BBT)
- b. Determination of hormonal excretion in urine
- c. Determination of hormonal level in plasma
- d. Ultrasound scanning
- e. Cervical mucus pattern

1. DETERMINATION OF BASAL BODY TEMPERATURE (BBT)

Measured during the mid-period of the menstrual cycle



In the morning by placing a thermometer in the rectum / vagina



Fall in basal temperature prior to ovulation ($\pm 0.3^{\circ}\text{C} - 0.5^{\circ}\text{C}$)



After ovulation $\longrightarrow$ $\uparrow$ increase in temperature due to **thermogenic effect of progesterone**

2. DETERMINATION OF HORMONAL EXCRETION IN URINE

$\uparrow$ Increase in excretion of metabolic end products of estrogen & progesterone



- **Estrogen End Products** : Estrone, Estriol, $17\text{-}\beta\text{-estradiol}$
- **Progesterone End Product** : Pregnanediol

3. DETERMINATION OF HORMONAL LEVEL IN PLASMA

- **At the Time of Ovulation :—**
 - a. FSH level $\downarrow$ decreases
 - b. LH level $\uparrow$ increases
 - c. Estrogen level $\uparrow$ increases
- **After Ovulation :—**
 - Progesterone level $\uparrow$ increases

4. ULTRASOUND SCANNING

- The process of follicular growth, maturation, and rupture can be directly observed via **ultrasound scanning**.

5. CERVICAL MUCUS PATTERN

Examined under a microscope → shows a characteristic **Fern Pattern**



Pattern **disappears** after ovulation